

MAKERERE



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MARKOV CHAIN MONTE CARLO METHODS

BY

GROUP 4

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1. HISTORICAL BACKGROUND OF MONTE CARLO ALGORITHM

1.1 Origins and Early Development (Genesis of the Monte Carlo Algorithm for Statistical Mechanics)

The Monte Carlo algorithm was developed between 1949 and 1952 at Los Alamos National Laboratory during a period of intense research related to the hydrogen bomb, their main mission was to understand the physics needed to build the **hydrogen bomb (H-bomb)**. Their need to understand the physics of radiation and matter interactions under extreme conditions (how explosion/implosion/explosion would evolve) drove the search for new computational methods. The first electronic computers (e.g., SEAC, JOHNNIAC, MANIAC- Mathematical Analyzer Numerical Integrator, and Calculator) were just becoming operational, enabling more complex calculations than previously possible. The method's development involved scientists such as Stan Ulam (who originated the basic idea and the name), John von Neumann, Nick Metropolis, Edward Teller, Arianna Rosenbluth, and Marshall N. Rosenbluth.

The main challenge was to calculate equilibrium properties (like the equation of state) for many-body systems, which was analytically intractable. Early attempts at molecular dynamics were limited by computational power. Edward Teller suggested leveraging statistical mechanics to focus on ensemble averages rather than detailed trajectories/configurations, leading to the Monte Carlo approach (Metropolis et al., 1953).

The Monte Carlo Algorithm

Instead of evaluating all possible trajectories or configurations, the Monte Carlo method uses random sampling to estimate ensemble averages. This is especially powerful for high-dimensional integrals, where traditional methods are computationally prohibitive/not feasible.

The algorithm involves making random "pseudo-moves" in configuration space and accepting or rejecting them based on energy changes and probability criteria (e.g., accepting moves that lower energy, or accepting higher-energy moves with a probability proportional to $\exp(-\Delta E/T)$). i.e. Instead of generating an entirely new configuration, the algorithm starts from a valid configuration and proposes a small change (**pseudo-move**), such as moving one particle by a short distance. The purpose of this move is not to simulate the actual motion of particles through time, but to generate another possible configuration of the system in configuration space.

After each trial move, the algorithm calculates the **energy difference** (ΔE) between the new and

current configurations and applies an **acceptance rule**:

- ▶ If the trial move **reduces the system's energy** ($\Delta E < 0$), the move is **always accepted**.
- ▶ If the trial move **increases the system's energy** ($\Delta E > 0$), it is accepted with probability $P = e^{-\Delta E/T}$ where: ΔE = increase in energy, and T = temperature.

A **random number** is generated and compared with this probability to determine whether the move is accepted or rejected. If the move is accepted, the new configuration becomes the current configuration. If it is rejected, the system remains in its previous configuration. This process is repeated many times. **the Monte Carlo algorithm always converges to the correct equilibrium distribution**, regardless of the starting state i.e. **Canonical distribution** $P_{\square} \propto e^{-E_{\square}/T}$. The probability after the next move is

$$P_{\nu}^{N+1} = \sum_{E_{\nu'} > E_{\nu}} P_{\nu'}^N T_{\nu'\nu} + \sum_{E_{\nu'} < E_{\nu}} P_{\nu'}^N T_{\nu'\nu} e^{-|\Delta E|/T} + P_{\nu}^N \sum_{E_{\nu'} > E_{\nu}} T_{\nu\nu'} (1 - e^{-|\Delta E|/T}).$$

Where; $\sum_{E_{\nu'} > E_{\nu}} P_{\nu'}^N T_{\nu'\nu}$ represents the probability flowing **into** state ν from **higher-energy states**. Since these moves lower the energy, they are **always accepted**,

$\sum_{E_{\nu'} < E_{\nu}} P_{\nu'}^N T_{\nu'\nu} e^{-|\Delta E|/T}$ represents probability entering state ν from **lower-energy states**. These moves increase the energy and are accepted only with probability $e^{-|\Delta E|/T}$.

$P_{\nu}^N \sum_{E_{\nu'} > E_{\nu}} T_{\nu\nu'} (1 - e^{-|\Delta E|/T})$ is the probability of **remaining in the current state** because a proposed move was rejected (Metropolis et al. 1953, pg 4).

Statistical/Methodological problem that existed and gap solved by Metropolis Algorithm (1953)

The main statistical problem was the inability to compute ensemble averages for many-body systems using traditional analytic or direct simulation methods. Specifically:

- ▶ **High-dimensional integrals:** Calculating ensemble averages for systems with hundreds of particles required evaluating integrals in extremely high-dimensional spaces ($d \times N$ dimensions, where d is the number of spatial dimensions and N is the number of particles).
- ▶ **Inefficiency of direct simulation:** Straightforward molecular dynamics required very small-time steps and was computationally infeasible with the limited capabilities of early computers.
- ▶ **Sampling rare but important configurations:** Randomly placing particles often resulted in configurations with very low probability, making it difficult to sample the relevant parts of phase space efficiently.

The Monte Carlo algorithm addressed the gap by providing a practical and efficient way to sample the configuration space of many-body systems and compute ensemble averages: **Example:**

Using the Monte Carlo method, the team was able to calculate the equation of state for hard disks and spheres, and later for systems with Lennard-Jones potentials, revealing new insights into phase behavior that were previously inaccessible.

2. Monte Carlo Sampling Methods Using Markov Chains and Their Applications

Hastings (1970) presents a landmark generalization of the **Metropolis et al. (1953) algorithm**. Prior to this, the Metropolis method was primarily a physics tool for simulating molecules. Hastings recognized that the underlying idea of generating samples from a distribution using a Markov chain that only requires **ratios** of the target density was incredibly powerful for **high-dimensional statistical problems** where traditional numerical integration fails.

2.1 Statistical/Methodological problem that existed and gap solved by Metropolis-Hastings Algorithm (1970)

The statistical problem that motivated the development of Markov Chain Monte Carlo (MCMC) methods was the difficulty of evaluating expectations with respect to complex probability distributions arising in high-dimensional statistical problems. Hastings observed that although Monte Carlo methods are generally more efficient than conventional numerical integration methods for multidimensional problems, they require samples from the required probability distribution, which are often difficult to obtain directly (Hastings, 1970).

To address this problem, Hastings reviewed three existing approaches.

1. The **first approach, factorization**, decomposes the joint probability distribution into a sequence of conditional distributions,

$$p(x_1, x_2, \dots, x_n) = p(x_1) p(x_2 | x_1) \cdots p(x_n | x_1 \dots x_{n-1}),$$

thereby allowing successive sampling from each conditional distribution. However, Hastings noted that such factorization is often difficult or impossible for complex statistical models (Hastings, 1970, p. 98).

2. The **second approach, importance sampling**, replaces the difficult target distribution $p(x)$ with a simpler proposal distribution $q(x)$. The expectation is rewritten as

$$J = \int f(x)p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx,$$

where the ratio $w(x) = \frac{p(x)}{q(x)}$ is the importance weight used to correct for the discrepancy between the proposal and target distributions. Although this method enables sampling from an easier distribution, its performance deteriorates when the proposal distribution differs sub-

stantially from the target distribution, resulting in highly variable importance weights (Hastings, 1970, p. 98).

3. The **third approach**, referred to by Hastings as the **simulation approach**, generates observations from another distribution, say $q(y)$, and transforms them into samples from the required distribution through a mapping $x = g(y)$. Alternatively, observations satisfying a specified condition, $h(y) = h_0$, are retained while the remaining observations are discarded. Hastings observed that this approach becomes computationally inefficient when the probability of obtaining acceptable observations is very small, since a large proportion of the simulated observations are rejected (Hastings, 1970)

The limitations of these three approaches highlighted the need for a more general simulation method capable of generating samples from complex probability distributions without requiring direct sampling, exact factorization, or unstable importance weights. This statistical challenge motivated Hastings to develop a Markov chain-based sampling method, which forms the foundation of the Metropolis-Hastings algorithm.

Integration Problems in Bayesian Inference

Many applications of Markov Chain Monte Carlo methods are problems that arise in Bayesian inference. From a Bayesian perspective, in a statistical model both the observables and the parameters are random. Given data $x = \{x_1, \dots, x_n\}$ and parameters θ , the relationship is expressed via the likelihood $f(x_1, \dots, x_n | \theta)$ and a prior density $f_\theta(\theta)$. The joint distribution is

$$f_{x,\theta}(x, \theta) = f_{x|\theta}(x_1, \dots, x_n | \theta) f_\theta(\theta).$$

By Bayes' theorem, the posterior distribution of θ given the data is

$$f_{\theta|x}(\theta | x) = \frac{f_{x|\theta}(x | \theta) f_\theta(\theta)}{\int f_{x|\theta}(x | \theta) f_\theta(\theta) d\theta}.$$

The conditional expectation of a function $g(\theta)$ with respect to the posterior is

$$E[g(\theta | x)] = \int g(\theta) f_{\theta|x}(\theta) d\theta = \frac{\int g(\theta) f_{x|\theta}(x | \theta) f_\theta(\theta) d\theta}{\int f_{x|\theta}(x | \theta) f_\theta(\theta) d\theta}.$$

More generally, one often needs to compute

$$E[g(Y)] = \frac{\int g(t) \pi(t) dt}{\int \pi(t) dt},$$

where $\pi(\cdot)$ is known only up to a multiplicative constant (e.g., an unnormalized posterior or likelihood). The normalising constant in the denominator may be intractable, the crucial advantage of MCMC is that it can handle this ratio without needing the normalizing constant, only the *ratio* of densities is required in the acceptance step. (Maria L. Rozzi, 2019)

2.2 The Metropolis-Hastings Algorithm (1970)

The Metropolis-Hastings algorithm's core idea is to construct a Markov chain $\{X_t | t = 0, 1, 2, \dots\}$ such that its stationary distribution is the target distribution. At each step, a candidate state Y is drawn from a proposal distribution $g(\cdot | X_t)$ that may depend on the current state X_t . The candidate is then accepted with a certain probability; if accepted, the chain moves to Y , otherwise it stays at X_t .

The proposal distribution must be chosen to ensure that the resulting chain is irreducible, aperiodic, and positive recurrent, conditions that guarantee convergence to a unique stationary distribution.

Let $P = \{p_{ij}\}$ be the transition matrix of an irreducible Markov chain, where;
 $p_{ij} = P\{X(t+1) = j | X(t) = i\}$. A fundamental result in Markov chain theory establishes that if π is the stationary distribution satisfying $\pi = \pi P$, then the estimator $(E_\pi[g(X)])$ converges to the true mean square and is asymptotically normally distributed.

Construction of the Transition Matrix

To guarantee stationarity, Hastings constructs P to satisfy the **detailed balance** (reversibility) condition: $\pi_i p_{ij} = \pi_j p_{ji}$ for all i, j .

If this holds, summing over i yields $\sum_i \pi_i p_{ij} = \pi_j \sum_i p_{ji} = \pi_j$. Hence, $\pi = \pi P$ is automatically satisfied. The irreducibility of the resulting chain must be verified separately for each specific application. Hastings assumes the transition probability from state i to state j can be expressed as the product of a proposal probability q_{ij} and an acceptance probability α_{ij} : $p_{ij} = q_{ij} \alpha_{ij}$ ($i \neq j$). The probability of remaining in the current state is consequently defined as: $p_{ii} = 1 - \sum_{j \neq i} p_{ij}$. Substituting into the reversibility condition yields the fundamental constraint on the acceptance probabilities: $\pi_i q_{ij} \alpha_{ij} = \pi_j q_{ji} \alpha_{ji}$.

MCMC General Acceptance Probability

Hastings presents a general solution for α_{ij} that automatically satisfies **detailed balance** (reversibility) condition. He sets:

$$\alpha_{ij} = \frac{s_{ij}}{1 + \frac{\pi_i q_{ij}}{\pi_j q_{ji}}},$$

where s_{ij} is **any symmetric function** of i and j (i.e., $s_{ij} = s_{ji}$) chosen such that $0 \leq \alpha_{ij} \leq 1$. Hastings identifies two notable special cases for the symmetric function s_{ij} :

Case 1: Metropolis's Method (1953)

Let

$$s_{ij}^{(M)} = 1 + \min \left(\frac{\pi_i q_{ij}}{\pi_j q_{ji}}, \frac{\pi_j q_{ji}}{\pi_i q_{ij}} \right)$$
$$\alpha_{ij}^{(M)} = \min \left(1, \frac{\pi_j q_{ji}}{\pi_i q_{ij}} \right).$$

Test Ratio defined as: $\frac{\pi_j q_{ji}}{\pi_i q_{ij}}$. If the proposal matrix is **symmetric** (i.e., $q_{ij} = q_{ji}$), this further reduces to:

$$\alpha_{ij}^{(M)} = \min \left(1, \frac{\pi_j}{\pi_i} \right).$$

Interpretation: The candidate state j is accepted with certainty if it has higher probability density than the current state i ; otherwise, it is accepted with probability equal to the ratio of the densities.

Case 2: Barker's Method (1965)

Let

$$s_{ij}^{(B)} = 1$$
$$\alpha_{ij}^{(B)} = \frac{1}{1 + \frac{\pi_i q_{ij}}{\pi_j q_{ji}}} = \frac{\pi_j q_{ji}}{\pi_i q_{ij} + \pi_j q_{ji}}.$$

With a symmetric proposal, this reduces to:

$$\alpha_{ij}^{(B)} = \frac{\pi_j}{\pi_i + \pi_j}.$$

3. Mathematical Development

3.1 Re-derivation of the Estimator

The objective of Monte Carlo methods is to estimate the expected value of a function with respect to a target probability distribution using random sampling techniques (Metropolis et al., 1953; Rizzo, 2019). Let X be a random variable with target probability distribution $\pi(x)$, and let $h(X)$ denote a function of interest. The parameter of interest is the expected value of $h(X)$ with respect to $\pi(x)$ and is defined as

$$\theta = E_{\pi}[h(X)] \quad (1)$$

By the definition of mathematical expectation, if X is a discrete random variable,

$$\theta = \sum_x h(x)\pi(x), \quad (2)$$

whereas if X is a continuous random variable,

$$\theta = \int h(x)\pi(x) dx. \quad (3)$$

In many practical applications, particularly in Bayesian inference and statistical mechanics, this expectation cannot be evaluated analytically because the integral is intractable or because the normalizing constant of the target distribution is unknown (Metropolis et al., 1953; Rizzo, 2019)

Suppose that it is possible to generate a sample $X_1, X_2, \dots, X_N$

from the target distribution $\pi(x)$. For each sampled value, the corresponding function values are

$$h(X_1), h(X_2), \dots, h(X_N).$$

The arithmetic mean of these function values is

$$\frac{h(X_1) + h(X_2) + \dots + h(X_N)}{N},$$

which may be written compactly as

$$\frac{1}{N} \sum_{i=1}^N h(X_i). \quad (4)$$

According to the Law of Large Numbers, the sample average converges to the population expectation as the sample size becomes sufficiently large (Robert & Casella, 2004)

$$\frac{1}{N} \sum_{i=1}^N h(X_i) \longrightarrow E_{\pi}[h(X)] = \theta. \quad (5)$$

Hence, the Monte Carlo estimator of θ is

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N h(X_i) \quad (6)$$

For Markov Chain Monte Carlo (MCMC) methods, the generated samples are not independent because they are obtained from a Markov chain. However, provided that the Markov chain is ergodic, the Ergodic Theorem guarantees that the sample average still converges to the desired expectation (Tierney, 1994) Consequently, the Monte Carlo estimator remains valid after the Markov chain has converged to its stationary distribution.

3.2 Re-derivation of the Target (Posterior) Distribution

In Bayesian statistics, inference is based on the posterior distribution, which combines prior knowledge about the unknown parameter with the information contained in the observed data through Bayes' theorem (Gelman et al., 2013).

Let θ denote the unknown parameter of interest and let y represent the observed data. Bayes' theorem is given by

$$P(\theta|y) = \frac{P(y|\theta)P(\theta)}{P(y)}, \quad (7)$$

where

- $P(\theta|y)$ is the posterior distribution,
- $P(y|\theta)$ is the likelihood function,

- $P(\theta)$ is the prior distribution,
- $P(y)$ is the marginal likelihood (evidence).

The marginal likelihood is obtained by integrating the joint distribution over all possible values of the parameter,

$$P(y) = \int P(y|\theta)P(\theta) d\theta. \quad (8)$$

Substituting the expression for the marginal likelihood into Bayes' theorem gives

$$P(\theta|y) = \frac{P(y|\theta)P(\theta)}{\int P(y|\theta)P(\theta) d\theta}. \quad (9)$$

The denominator is a normalizing constant that does not depend on θ . Since the Metropolis–Hastings algorithm compares posterior probabilities through ratios, this constant cancels and therefore does not need to be evaluated explicitly (Hastings, 1970).

Consequently,

$$\boxed{P(\theta|y) \propto P(y|\theta)P(\theta)} \quad (10)$$

Using standard statistical notation,

$$\boxed{\pi(\theta|y) \propto L(\theta|y)p(\theta)} \quad (11)$$

The above expression defines the target posterior distribution from which the Metropolis–Hastings algorithm generates samples (Gelman et al., 2013; Robert & Casella, 2004).

3.3 Re-derivation of the Objective Function

Unlike classical estimation methods that seek to maximise or minimise a specific criterion (such as a log-likelihood or a sum of squares), the Metropolis–Hastings algorithm does not have a single objective function to optimise. Instead, its primary goal is to construct a Markov chain whose stationary distribution matches the target distribution $\pi(x)$ (Hastings, 1970). This is achieved by designing the transition kernel of the chain so that it satisfies the *detailed balance* (or reversibility) condition:

$$\pi(x)K(x, x') = \pi(x')K(x', x), \quad (12)$$

where

- $\pi(x)$ is the target distribution,
- $K(x, x')$ is the probability of moving from state x to state x' in one step.

The transition kernel is built from two parts: a proposal distribution $q(x'|x)$ that suggests a candidate move, and an acceptance probability $\alpha(x, x')$ that decides whether to take that move. Thus,

$$K(x, x') = q(x'|x)\alpha(x, x'). \quad (13)$$

Similarly, the reverse transition is

$$K(x', x) = q(x|x')\alpha(x', x). \quad (14)$$

Substituting these into the detailed balance equation yields

$$\pi(x)q(x'|x)\alpha(x, x') = \pi(x')q(x|x')\alpha(x', x). \quad (15)$$

To find an acceptance probability that always lies between 0 and 1, Hastings (?) proposed the now-standard form:

$$\alpha(x, x') = \min \left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)} \right) \quad (16)$$

This expression is the **Metropolis–Hastings acceptance probability**. It effectively says: if the proposed state is more probable under the target (or the proposal is easier to reverse), we always accept it; otherwise, we accept it with a probability proportional to the ratio of the target densities and the proposal densities. By construction, this choice guarantees that the detailed balance condition holds, and therefore $\pi(x)$ is the stationary distribution of the Markov chain (Hastings, 1970; Metropolis et al., 1953).

Intuitive interpretation. The ratio inside the minimum compares how likely the proposed state is relative to the current state, while also correcting for any asymmetry in the proposal mechanism. If the proposal distribution is symmetric (i.e., $q(x'|x) = q(x|x')$), this formula simplifies to the original Metropolis ratio $\pi(x')/\pi(x)$. In that case, we always move to states with higher target density, and only occasionally move to lower-density states—thus the chain naturally spends more time in regions of high probability, mirroring the target distribution.

3.4 Re-explanation of the Optimization (Sampling) Procedure

The Metropolis–Hastings algorithm is an iterative sampling technique used when direct sampling from the target distribution is infeasible. Rather than searching for a single optimal value, it simulates a random walk that gradually converges to the target distribution. The steps below describe this process in a practical, step by step manner, as commonly presented in computational statistics textbooks (Rizzo, 2019; Robert & Casella, 2004).

1. Initialization

Choose a starting state $x^{(0)}$. This can be any value in the parameter space, though starting near the centre of the target distribution can reduce the number of iterations needed to reach equilibrium.

2. Propose a Candidate State

At the current iteration t , with state $x^{(t)}$, generate a candidate x^* from a proposal distribution $q(x^* | x^{(t)})$. This distribution is chosen by the user and should be easy to sample from. Common choices include a normal distribution centred at $x^{(t)}$ (random walk) or a fixed independent distribution.

3. Compute the Acceptance Probability

Calculate

$$\alpha = \min \left(1, \frac{\pi(x^*)q(x^{(t)}|x^*)}{\pi(x^{(t)})q(x^*|x^{(t)})} \right). \quad (17)$$

This number summarises how favourable the proposed move is. If α is large, the candidate is very plausible relative to the current state; if it is small, the move is unlikely to be beneficial.

4. Accept or Reject

Draw a uniform random number $u \sim U(0, 1)$. If $u \leq \alpha$, we accept the candidate and set

$x^{(t+1)} = x^*$. Otherwise, we reject it and stay put: $x^{(t+1)} = x^{(t)}$. Notice that even when we reject, the current state is counted again in the sample—this is essential for the chain to correctly represent the stationary distribution.

5. Iterate

Repeat steps 2–4 many times. The first portion of the chain (the “burn-in”) is discarded because it may still be influenced by the arbitrary starting value. After the burn-in, the chain is considered to have converged, and subsequent states are treated as dependent samples from $\pi(x)$.

6. Estimate Quantities of Interest

After convergence, for any function $h(x)$, the expectation under π is estimated by the sample average:

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N h(x_i), \quad (18)$$

where $x_1, \dots, x_N$ are the post-burn-in samples. This is the same Monte Carlo estimator presented earlier, now applied to correlated MCMC output.

The beauty of this procedure is that it requires only the ability to evaluate the target density up to a constant (since the normalising factor cancels in the ratio). This makes MCMC exceptionally flexible for high-dimensional Bayesian models and complex physical systems (Hastings, 1970; Metropolis et al., 1953)

4. Statistical Assumptions

The validity of the Metropolis–Hastings algorithm depends on several assumptions that ensure the Markov chain converges to the desired target distribution and produces reliable Monte Carlo estimates.

i. Markov Property

The algorithm assumes that the sequence of generated samples forms a Markov chain. Consequently, the probability of moving to the next state depends only on the current state and not on the previous history of the chain. Mathematically,

$$P(X_{t+1} \mid X_t, \dots, X_0) = P(X_{t+1} \mid X_t). \quad (19)$$

This property simplifies the sampling process and defines the stochastic dependence structure of the chain.

ii. Irreducibility

The Markov chain must be irreducible, meaning that every state in the support of the target distribution can be reached from every other state after a finite number of transitions. This ensures that the chain is capable of exploring the entire parameter space.

iii. Aperiodicity

The chain must not become trapped in deterministic cycles. Instead, it should be possible to revisit states at irregular intervals, allowing convergence to the target distribution.

iv. Detailed Balance

The transition probabilities must satisfy the detailed balance condition,

$$\pi(x)K(x, x') = \pi(x')K(x', x), \quad (20)$$

where $K(x, x')$ is the transition kernel and $\pi(x)$ is the target distribution. This condition guarantees that the probability flow from state x to state x' is equal to the reverse flow, thereby preserving the target distribution.

v. Stationary Distribution

The target distribution, $\pi(x)$, is assumed to be the stationary distribution of the Markov chain. Consequently, once the chain has converged, repeated sampling leaves the distribution unchanged, that is,

$$\pi(x) = \sum_{x'} \pi(x') K(x', x). \quad (21)$$

This property ensures that samples generated after convergence are representative of the target distribution.

vi. Ergodicity

The Markov chain must be ergodic. Under irreducibility and aperiodicity, the chain converges to the stationary distribution regardless of its initial starting point. As the number of iterations increases, sample averages converge to the corresponding expectations under the target distribution.

vii. Proposal Distribution

The proposal distribution, $q(x'|x)$, must assign positive probability to every region where the target distribution is positive. In other words,

$$q(x'|x) > 0 \quad \text{whenever} \quad \pi(x') > 0. \quad (22)$$

This condition prevents the chain from excluding regions of the parameter space that contribute to the target distribution.

5. Data Description

5.1 Source of Data

The data used in this study were obtained from the **Pima Indians Diabetes Database**, a well-established benchmark dataset that has been widely applied in statistical modelling and machine learning studies involving diabetes mellitus. The dataset was originally compiled by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and is publicly available through the UCI Machine Learning Repository (Blake & Merz, 1998).

The version used in this analysis is the `PimaIndiansDiabetes2` dataset, which was adapted for use in R by Friedrich Leisch and is distributed through the `mlbench` package (Leisch & Dimitriadou, 2024). The dataset consists exclusively of adult female participants of Pima Indian heritage, all of whom were at least 21 years of age at the time of data collection.

5.2 Sample Size

The dataset comprises **768 observations**, each representing an individual patient. For every patient, information was collected on nine variables, consisting of eight clinical characteristics and one diabetes diagnosis variable.

5.3 Variables

Table 1 presents the variables included in the dataset together with their descriptions and their roles in the present analysis.

Table 1: Description of study variables

Variable	Description	Role in this study
pregnant	Number of times the patient had been pregnant	Predictor
glucose	Plasma glucose concentration obtained from an oral glucose tolerance test (mg/dL)	Response variable
pressure	Diastolic blood pressure (mm Hg)	Predictor
triceps	Triceps skinfold thickness (mm)	Predictor
insulin	Two-hour serum insulin concentration (μ U/mL)	Predictor
mass	Body Mass Index (BMI) (kg/m^2)	Imputed but excluded from the final regression models
pedigree	Diabetes pedigree function, representing genetic predisposition to diabetes	Predictor
age	Age of the patient (years)	Predictor
diabetes	Diabetes diagnosis (positive/negative)	Used only for descriptive statistics; excluded from the regression models

5.4 Missing Data

An assessment of missing values was conducted prior to model fitting. Several variables contained biologically implausible zero values, which were treated as missing observations. These occurred primarily in the insulin and triceps skinfold thickness variables, with smaller proportions observed for glucose, blood pressure and BMI.

Table 2 summarises the extent of missingness for each affected variable.

Table 2: Extent of missing data

Variable	Percentage Missing	Approximate Number of Patients
Insulin	49.00%	≈ 376
Triceps skinfold thickness	30.00%	≈ 230
Blood pressure	4.56%	≈ 35
Body Mass Index (BMI)	1.43%	≈ 11
Glucose	0.65%	≈ 5

Insulin and triceps skinfold thickness exhibited substantial levels of missingness, whereas the proportions of missing observations in glucose, blood pressure and BMI were comparatively small.

Missing Data Mechanism

The missing values were assumed to be *Missing at Random* (MAR). Under this assumption, the probability of missingness depends on observed characteristics rather than on the unobserved values themselves. In this study, it was considered plausible that missingness in variables such as BMI and glucose was associated with observed patient characteristics, including age and pregnancy history, rather than the missing measurements. As discussed in Section 10, however, the MAR assumption cannot be verified empirically using the observed data alone and should ideally be justified based on knowledge of the data collection process rather than assumed during analysis.

Multiple Imputation

Missing values were handled using Multiple Imputation by Chained Equations (MICE) with predictive mean matching (PMM) as the imputation method (van Buuren & Groothuis-Oudshoorn, 2011). Twenty imputed datasets ($m = 20$) were generated using a maximum of 50 iterations to facilitate convergence. A random seed of 123 was specified to ensure reproducibility of the imputation process. Following imputation, the completed datasets contained no missing values.

5.5 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to examine the distribution of the variables, assess relationships among the predictors, and evaluate the assumptions underlying the regression models.

Univariate Distributions

The distributions of the continuous variables were examined using boxplots. Glucose was the only variable that did not exhibit any apparent outliers. In contrast, pregnancies, age, insulin, BMI, diabetes pedigree function and triceps skinfold thickness showed extreme observations primarily in the upper tails of their respective distributions. Blood pressure exhibited potential outliers at both the lower and upper extremes.

These observations were considered clinically plausible rather than data entry errors. Consequently, they were retained in the dataset to preserve the natural variability of the study population.

Relationships Among Variables

Pairwise relationships among the study variables were examined using a correlation matrix generated with the `ggpairs` function. This analysis also provided an informal assessment of the linearity assumption required for classical linear regression.

Three pairs of variables exhibited relatively strong positive correlations exceeding 0.50. Triceps skinfold thickness and BMI were the most strongly correlated ($r = 0.623$), followed by glucose and insulin ($r = 0.604$), and pregnancies and age ($r = 0.544$).

A broader pattern of moderate positive associations was also observed among glucose, insulin, BMI, triceps skinfold thickness and blood pressure, reflecting the expected clustering of metabolic risk factors. Age showed moderate positive correlations with most of the clinical variables, with correlation coefficients ranging from 0.233 to 0.329. BMI was moderately correlated with blood pressure ($r = 0.290$) and insulin ($r = 0.231$).

In contrast, the diabetes pedigree function displayed relatively weak associations with the remaining variables, with correlation coefficients of 0.136, 0.153 and 0.034 for glucose, BMI and age, respectively. Similarly, the number of pregnancies showed negligible correlations with insulin ($r = 0.054$), BMI ($r = 0.025$) and diabetes pedigree function ($r = -0.034$).

Overall, the correlation structure suggested moderate linear relationships among several predictors but did not indicate excessive dependence between variables.

Assessment of Multicollinearity

Multicollinearity was assessed using the Variance Inflation Factor (VIF) based on the classical linear regression model. All predictor variables produced VIF values below the commonly accepted threshold of 5, indicating that multicollinearity was not a significant concern. This finding was

consistent with the moderate pairwise correlations observed during the exploratory analysis.

Table 3: Variance Inflation Factor (VIF) Results

Variable	VIF
Pregnant	1.439169
Pressure	1.168024
Triceps	1.107006
Insulin	1.102213
Pedigree	1.031938
Age	1.612881

Assessment of Residual Normality

Residual normality was assessed using a normal quantile–quantile (Q–Q) plot obtained from the classical linear regression model. The residuals followed the reference line reasonably well, suggesting that the normality assumption was largely satisfied. However, slight departures from normality were evident in the tails of the distribution.

Assessment of Homoscedasticity

The assumption of constant error variance was evaluated using the Breusch–Pagan test (Breusch & Pagan, 1979). When the model was fitted using the original glucose measurements, the test produced a p-value below the 0.05 significance level, providing evidence of heteroscedasticity.

To address this issue, the response variable was transformed using the natural logarithm, and the model was refitted. The Breusch–Pagan test performed on the transformed model yielded a p-value greater than 0.05, indicating that the assumption of homoscedasticity was no longer violated. This suggests that the logarithmic transformation successfully stabilised the variance of the residuals.

6. How well do the assumptions hold for our dataset?

The assumptions underlying the Bayesian and Gibbs sampling approaches were evaluated using the Pima Indians Diabetes dataset as follows.

- ▶ **Markov Property.** The Markov property is satisfied by construction rather than through empirical assessment. Both sampling algorithms generate new states conditional only on the current state of the Markov chain. In the Stan implementation, new proposals are generated using the No-U-Turn Sampler (NUTS), which constructs Hamiltonian trajectories based on the current position of the chain. Similarly, the Gibbs sampler implemented through MCMCregress generates new parameter values from the corresponding full conditional distributions given the current values of the remaining parameters. Therefore, neither algorithm depends on previous iterations beyond the current state of the chain.
- ▶ **Irreducibility.** Irreducibility cannot be directly verified from a single simulation run; however, empirical diagnostics provide evidence that both samplers explored the posterior parameter space effectively. For the Gibbs sampler, the effective sample size (`effectiveSize(combined_chains)`) ranged from 75,805 to 83,107 out of 80,000 post-burn-in draws (four chains combined). For the Stan implementation, the median bulk effective sample size (n_{eff}) ranged from 6,396 to 8,874 out of 8,000 post-warm-up draws. These large effective sample sizes suggest good mixing and indicate that the chains were not practically restricted to a limited region of the posterior distribution.
- ▶ **Aperiodicity.** Aperiodicity was assessed through visual examination of trace plots (Section 3.3.2 for Stan and Section 3.4.3 for Gibbs). The trace plots for both sampling approaches displayed irregular fluctuations around stable posterior means, with no visible cyclic patterns or systematic oscillations across the six predictor parameters. These results provide evidence consistent with an aperiodic chain.
- ▶ **Detailed Balance.** Similar to the Markov property, detailed balance is a theoretical property established through the construction of the sampling algorithms rather than an assumption that can be empirically tested from the dataset. The Gibbs sampler satisfies detailed balance through its sequential updates from the exact full conditional distributions. In the Stan implementation, the Metropolis correction step within the NUTS algorithm ensures that the resulting transition kernel preserves the target posterior distribution.
- ▶ **Stationary Distribution.** The assumption that the chains converge to the target stationary distribution was strongly supported by the convergence diagnostics. The Gelman–Rubin po-

tential scale reduction factor ($\hat{R}$) was equal to 1.0 for all parameters in both the Stan/NUTS model (Section 3.3.5) and the Gibbs model (Section 3.4.4). In addition, the upper 95% confidence limits for $\hat{R}$ were also equal to 1.0. Trace plots showed no evidence of chain drift, trends, or separation between chains, providing strong evidence that both samplers converged to a common stationary distribution.

- **Ergodicity.** Evidence supporting ergodic behaviour was obtained from the autocorrelation diagnostics presented in Sections 3.3.4 and 3.4.5. For both sampling methods, autocorrelation values declined rapidly towards zero within approximately one to two lags across all predictor parameters. Combined with the large effective sample sizes, these results indicate that the generated posterior samples provide reliable estimates of posterior expectations and demonstrate satisfactory mixing behaviour.

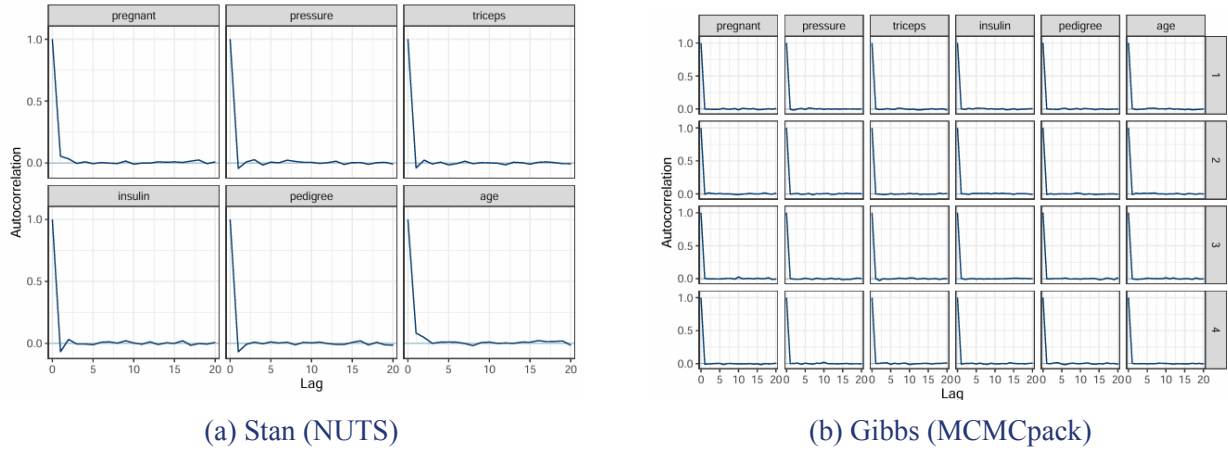


Figure 1: Autocorrelation comparison for the Stan/NUTS (a) and Gibbs (b) sampling models, showing rapid decay to zero within one to two lags for all six predictors consistent with the ergodicity assumption underlying valid MCMC inference

- **Proposal Distribution.** The proposal mechanisms for both sampling approaches were appropriate for the specified model structure. The Gibbs sampler draws directly from the exact full conditional distributions, which is possible because the model is based on conjugate normal linear regression assumptions. The Stan/NUTS implementation uses gradient-based Hamiltonian exploration to efficiently navigate the continuous posterior parameter space without requiring manual specification of a proposal distribution. Therefore, both approaches provide valid mechanisms for exploring the posterior distribution under the assumptions of the fitted model.

7. How does the method compare with contemporary alternatives (accuracy, interpretability, computational efficiency, robustness)?

Accuracy. Point estimates were highly consistent across all three methods. The posterior mean log scale coefficient estimate for the diabetes pedigree function, for instance, was 0.041 across all the three models (under the Stan, Gibbs sampler and the classical models). This convergence across estimation frameworks provides reassurance regarding the stability of the substantive findings.

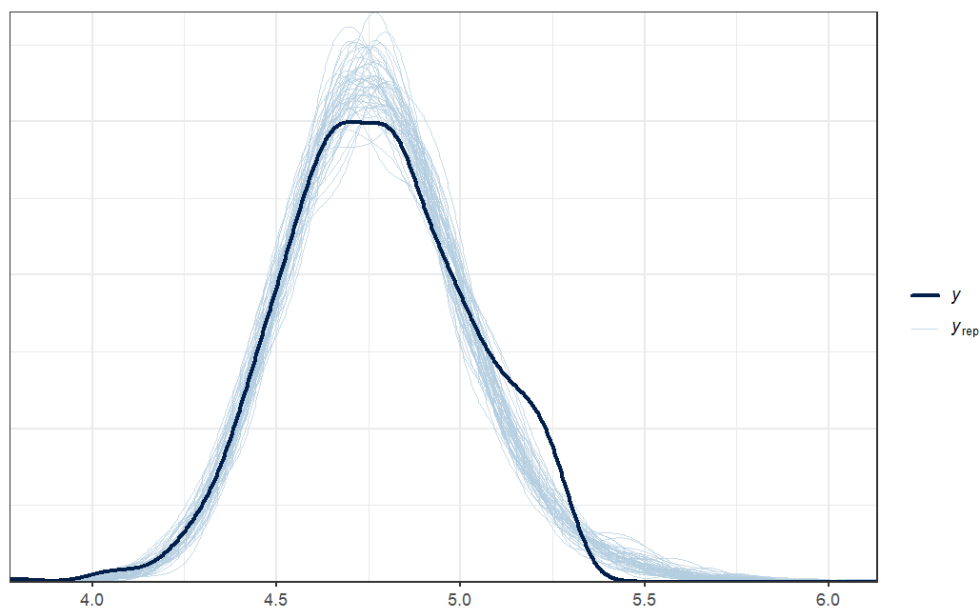


Figure 2: Posterior predictive check for the Bayesian (Stan) model, comparing observed glucose values (y) against posterior predictive replications (y_{rep}).

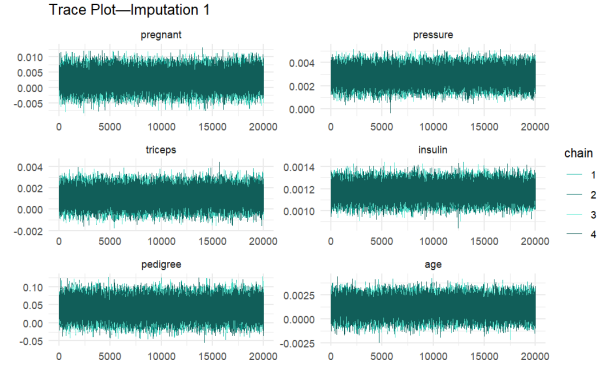
Interpretability The Bayesian and Gibbs approaches offer a distinct interpretive advantage over the classical framework by yielding a full posterior distribution rather than a single point estimate and standard error. This permits direct probabilistic statements (e.g. a 95% credible interval), which are arguably more intuitive for non-technical audiences than the corresponding frequentist confidence interval.

Robustness While Gibbs sampling matched Stan's accuracy and convergence properties on this conjugate linear model, it is considerably less robust to departures from this setting.

Stan provides substantially greater flexibility, including support for non-Gaussian likelihoods, hierarchical model structures, and non-conjugate priors, together with built-in diagnostic tooling for



(a) Stan (NUTS)



(b) Gibbs (MCMCpack)

Figure 3: Trace plots for the Bayesian (Stan/NUTS) and Gibbs sampling models. Both samplers exhibit stable, well-mixed chains with no visible drift across all six predictors, confirming that Gibbs sampling’s substantial speed advantage did not come at the cost of convergence quality.

detecting sampling pathologies that Gibbs sampling via MCMC pack does not provide.

Computational efficiency A marked difference was observed between the two MCMC implementations. The Gibbs sampler completed estimation in 22.52 seconds, compared to 313.20 seconds for the Stan/NUTS implementation, representing an approximate 13.7 fold improvement in speed. This efficiency gain is attributable to the fact that Gibbs sampling exploits the closed form conjugate full conditional distributions available under the normal linear model, whereas NUTS relies on a more computationally intensive Hamilton trajectory search that generalities to non conjugate settings.

8. Limitations of the original method and how later developments addressed them

The original **Metropolis-Hastings algorithm** requires the user to manually specify and tune an appropriate proposal distribution. Poorly selected proposal distributions can lead to low acceptance rates, increased autocorrelation among samples and inefficient exploration of the posterior distribution. Furthermore, the performance of the algorithm can deteriorate substantially as the dimensionality of the parameter space increases.

These limitations motivated the development of more advanced MCMC methods, including **Hamilton Monte Carlo (HMC)**, the **No-U-Turn Sampler (NUTS)** and **R-INLA a package for implementing Bayesian inference through integrated nested Laplace approximation models**. **INLA has demonstrated computational efficiency for latent Gaussian models (Woika et al., 2026)** These approaches utilise gradient information to generate more efficient proposals and automatically adapt key sampling parameters during the warm-up phase. This adaptive sampling strategy was employed in the Stan implementation presented in section 3.3, where the first 2000 of the 4000 iterations for each chain were allocated to warm-up.

The main limitation of the **Gibbs sampler** implemented in this study is its reliance on conjugate priors and analytically tractable full conditional distributions. This restricts its direct application to models where these conditional distributions can be derived in closed form. As a result, applying Gibbs sampling to non-Gaussian outcomes or more complex hierarchical models often requires additional derivations or extensions to the standard algorithm. Furthermore, Gibbs sampling may exhibit slow mixing when strong correlations exist among parameters in the posterior distribution. However, this was not a substantial issue in our analysis due to the relatively low correlation among predictor variables and the resulting stable posterior sampling behaviour.

A shared limitation of both the original **Metropolis-Hastings algorithm** and **classical Gibbs sampling** is the lack of automated diagnostic and model evaluation tools available in modern Bayesian software frameworks. In particular, they do not provide integrated posterior predictive checking, model comparison procedures, or automated detection of sampling issues comparable to tools available within the Stan ecosystem, such as `pp_check()`, `loo()`, and divergence diagnostics. These developments address important practical challenges in applied Bayesian modelling and have contributed to the widespread adoption of Stan, despite its higher computational requirements compared with simpler MCMC approaches.

9. Methodological improvements expected by reviewers today

If Hastings' (1970) paper were submitted to a leading statistics journal today, several issues would likely be raised by reviewers. Our review captures all these methodological improvements.

1. Single-chain evidence of convergence is insufficient.

Hastings demonstrates the performance of his algorithm using individual simulation runs. For example, in the normal sampling example, he uses $N = 1000$ and $L = 25$ and reports a reasonable estimate and standard deviation. However, modern MCMC analysis requires stronger evidence of convergence. A reviewer today would expect multiple independent chains with dispersed starting values. They would also expect formal convergence diagnostics, such as the Gelman–Rubin potential scale reduction factor ($\hat{R}$). Although this diagnostic was developed much later and could not have been included in the original paper, its absence would be considered a limitation today. A single chain that produces plausible results does not confirm that the chain has fully explored the target distribution or avoided being trapped in a local region.

2. No systematic approach for selecting the proposal distribution.

Hastings shows that the performance of the algorithm depends on the random-walk proposal step size Δ . He reports that moderate values between 0.2 and 1.8 produce reasonable results, while extreme values lead to poorer estimates. However, the paper does not provide a principled method for selecting this value. A modern reviewer would expect an adaptive procedure that selects the proposal scale based on a target acceptance rate. Alternatively, a systematic sensitivity analysis over different proposal values would be expected, including measures such as acceptance rates and effective sample sizes.

3. The estimation of variance under correlated samples remains approximate.

Hastings recognises that estimating the uncertainty of Monte Carlo averages from correlated Markov chain samples is challenging. He proposes two approximate solutions: estimating the spectral density at zero frequency and using a simpler batch-means estimator. However, these approaches are not evaluated against a known benchmark, such as repeated independent simulations of the entire chain. A modern reviewer would expect validation of these variance estimators. They would also expect reporting of effective sample sizes, which are now standard measures in MCMC analysis.

4. Convergence is established asymptotically, but convergence rates are not discussed.

Hastings proves that the proposed Markov chain converges to the target distribution as the number of iterations increases. However, the paper does not provide information on the speed of convergence. Modern theoretical work on MCMC often considers convergence rates through concepts such as geometric ergodicity and spectral gaps. A contemporary reviewer would expect either a convergence-rate result or a discussion of the conditions required to obtain such results.

5. Lack of comparison with alternative methods.

The examples presented in the paper, including the normal distribution example and the random orthogonal matrix example, are evaluated independently without comparison to alternative approaches. For example, where direct sampling methods are available, these could have been used as benchmarks. A modern reviewer would expect comparisons with simpler or existing methods to demonstrate the additional value of the Markov chain approach.

6. Limited reproducibility by current standards.

Hastings reports simulation results obtained using an IBM 7094 II computer. However, the paper does not provide the source code or sufficient implementation details to reproduce the reported numerical results exactly. Modern statistical research places greater emphasis on reproducibility. A current submission would typically be expected to provide the implementation code, random seeds, and complete details of the simulation procedure.

10. Practical implications of MCMC for policy, research or industry in Statistics

- ▶ **Methodological choice in survey and health research contexts** The Bayesian framework provides an important advantage in applied research settings by allowing uncertainty to be represented through full posterior distributions rather than decisions based solely on binary significance thresholds. This is particularly valuable when communicating findings to policy.
- ▶ **Computationally constrained research settings:** The observed approximately nine-fold reduction in runtime achieved by Gibbs sampling compared with Stan/NUTS for the conjugate linear model represents a practical consideration in settings where computational resources or turnaround time are limited, such as rapid-cycle policy analysis. However, this computational advantage is closely linked to the simplicity and conjugate structure of the model. For more complex applications, including multilevel or hierarchical models, the greater flexibility and broader modelling capabilities provided by Stan may justify its higher computational requirements.

11. Circumstances in which MCMC is recommended over competing approaches

Based on the empirical findings in our analysis, we would recommend the following:

- ▶ **Gibbs sampling (MCMCpack)** is recommended where the model is a standard conjugate Gaussian linear regression, where computational speed and rapid iteration are priorities, and where posterior predictive checking and model extension beyond linear regression are not required.
- ▶ **Bayesian regression via Stan/NUTS** is recommended where the analysis requires extension beyond the linear-Gaussian case (e.g. generalised linear models, hierarchical or multilevel structures, or non-conjugate priors), or where built-in diagnostic tools for model validation are considered necessary.
- ▶ **Classical (frequentist) regression** remains an appropriate and defensible choice where a fast, familiar baseline is required, particularly for communication to audiences less familiar with Bayesian inference, and where the sample size is sufficiently large that point estimates from all three approaches converge, as observed in the present analysis.

Given that the classical, Bayesian, and Gibbs approaches produced materially consistent point estimates in this instance, the choice among them should be guided primarily by the analytical requirements beyond the point estimate itself namely:

1. the need for full posterior uncertainty and diagnostic infrastructure (favouring Stan),
2. computational efficiency for a simple conjugate model (favouring Gibbs), or interpretive simplicity for a rapid baseline (favouring classical estimation).

12. Critical Evaluation of Hastings (1970) Algorithm

12.1 Theoretical Strengths

► Rigorous Mathematical Foundation

The Metropolis-Hastings algorithm rests on an elegant theoretical framework that guarantees convergence to the target distribution. The algorithm satisfies the **detailed balance (reversibility)** condition:

$$\pi(x)K(x, x') = \pi(x')K(x', x)$$

This property ensures that the probability flow between any two states is equal in both directions, making $\pi(x)$ the stationary distribution of the Markov chain.

The acceptance probability derived by Hastings:

$$\alpha(x, x') = \min \left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)} \right)$$

represents a remarkably elegant solution, it automatically satisfies detailed balance without requiring the normalizing constant of the target distribution. This means the algorithm works with **unnormalized densities**, a property that makes it invaluable for Bayesian inference where the marginal likelihood is often intractable.

► The H-Theorem Proof

Marshall Rosenbluth's proof of the H-theorem established that deviations from the canonical distribution decay exponentially. The evolution equation:

$$P_{\nu}^{N+1} = \sum_{E_{\nu'} > E_{\nu}} P_{\nu'}^N T_{\nu'\nu} + \sum_{E_{\nu'} < E_{\nu}} P_{\nu'}^N T_{\nu'\nu} e^{-|\Delta E|/T} + P_{\nu}^N \sum_{E_{\nu'} > E_{\nu}} T_{\nu\nu'} (1 - e^{-|\Delta E|/T})$$

demonstrates that all eigenvalues of the transition operator fall in the range $1 > \lambda > 0$, with only the canonical distribution having $\lambda = 1$.

This theoretical guarantee of convergence is a major strength—unlike many numerical methods, the algorithm will eventually converge to the correct answer regardless of starting state.

► Built-in Error Assessment: Hastings provided the Block Method (Equation 2, Pages 3–4)

to estimate the variance of the sample mean without requiring independence. This was a massive step up from the physics literature, which often ignored the correlation entirely.

- ▶ **Conceptual Simplicity:** The algorithm is shockingly simple: propose, accept or reject, repeat. It transforms a global sampling problem into a sequence of local binary decisions, making it trivial to implement on a computer.

12.2 Methodological Weaknesses

- ▶ **Arbitrary Proposal Selection:** The algorithm's performance critically depends on the choice of proposal distribution, yet the original formulation provides no systematic guidance for this selection. Hastings acknowledged:

"The maximum displacement α must be chosen with some care; if too large, most moves will be forbidden, and if too small, the configuration will not change enough."

The subjective nature of proposal tuning remains a fundamental weakness. What works for one problem may fail entirely for another.

- ▶ **Convergence Assessment Challenges:** While the algorithm guarantees asymptotic convergence, determining **when** convergence has occurred in practice remains problematic.

Modern practice relies on multiple diagnostic measures, each with limitations:

Gelman-Rubin $\hat{R}$

- Requires multiple independent chains
- May fail for multi-modal distributions
- Can give false convergence for slow-mixing chains
- Assumes approximate normality of posterior

Effective Sample Size

- Cannot detect poor mixing in tails
- Highly sensitive to autocorrelation estimation
- May overestimate effective samples for multimodal distributions

Trace Plots

- Highly subjective visual assessment
- Can miss subtle mixing problems
- May appear converged while failing to explore all modes

- ▶ **The Burn-in Problem:** The requirement to discard initial samples (burn-in) is theoretically justified—the chain must reach its stationary distribution. However, in practice, determining burn-in length is **ad hoc and subjective**.

Hastings recommended choosing the initial state “in a region of high probability,” but this is often difficult when the target distribution is the object of inference.

- ▶ **Correlated Samples:** Hastings openly admits that the generated samples are correlated. This “memory” dramatically reduces the Effective Sample Size (ESS) relative to the number of iterations N . While he provides a fix (the Block Method), the estimation of variance remains less efficient than with i.i.d. samples.
- ▶ **Sensitive to Initialization:** Although asymptotically the starting point doesn’t matter, in practice the initial state heavily biases early samples (burn-in). Hastings recommends starting near the mode, but if the mode is unknown, the chain wastes massive compute in burn-in/warm up.

12.3 Computational Challenges

- ▶ **Burn-in Determination:** Deciding how many initial samples to discard is purely heuristic. Discarding too few biases the estimate; discarding too many wastes compute. Hastings acknowledges this but offers no diagnostic tool.
- ▶ **Dimensionality Curse:** The Metropolis-Hastings algorithm scales poorly with increasing dimension. For a random-walk Metropolis with Gaussian proposals, the optimal acceptance rate decreases as dimension increases: This means the algorithm spends an increasing proportion of time rejecting proposals as dimension grows. The Pima dataset (6 predictors) was within the feasible range, but for high-dimensional problems (hundreds or thousands of parameters), standard Metropolis-Hastings becomes impractical.
- ▶ **Autocorrelation and Dependence:** Unlike independent Monte Carlo sampling, MCMC produces correlated samples. This has two consequences:

Reduced effective sample size:

For an autocorrelation ρ at lag 1, the variance inflation factor is approximately $(1+\rho)/(1-\rho)$. For $\rho = 0.9$, this is a 19-fold reduction in effective sample size.

Difficult variance estimation:

Hastings proposed variance estimators based on spectral density or batch means, but these are approximate:

$$\text{var}(\bar{Y}) \simeq \frac{2\pi g(0)}{N}$$

The batch/block means estimator:

$$\delta_{\bar{Y}}^2 = \frac{\sum_{i=1}^L (\bar{Y}_i - \bar{Y})^2}{L(L-1)}$$

requires choosing the batch size K , another arbitrary tuning parameter.

► Handling Multi-Modal Distributions

The algorithm can fail catastrophically when the target distribution has multiple modes separated by low-probability regions. Consider a target with two modes at μ_1 and μ_2 separated by a valley. If the proposal variance is too small, the chain becomes trapped in one mode; if too large, most proposals are rejected. Neither situation yields correct inference.

► Computational Cost

The empirical comparison from the Pima analysis reveals the cost of sophisticated sampling:

Method	Time	Steps per Iteration
Gibbs	23.44s	Exact conditional draws
Stan/NUTS	313.20s	Gradient + leapfrog integration

The 13.7x speed advantage of Gibbs comes at the cost of requiring conjugate priors and closed-form conditionals, a restriction that severely limits model flexibility.

12.4 Situations Where the Method Fails

Based directly on Hastings' caveats (Pages 2, 5, and 9) and standard MCMC theory:

► Poorly Behaved Posteriors

High Correlation

High correlation between parameters creates narrow ridges in the posterior. Random-walk Metropolis struggles because proposals along the ridge are more likely than proposals perpendicular to it. For a bivariate normal with correlation $\rho = 0.95$, the effective sample size can be reduced by a factor of 20 relative to independent sampling.

Heavy-Tailed Distributions

Heavy-tailed distributions can cause the chain to produce occasional extreme values, leading to:

- Slow convergence of ergodic averages
- Highly variable estimates
- Difficulty determining sufficient iteration count

► Poor Practical Implementation

Insufficient Iterations

Insufficient iterations is a common failure mode. Hastings noted:

”If the sample size is not large enough, important regions of the sample space may be inadequately represented.”

Poor Starting Values

Poor starting values in low-probability regions can lead to:

- Excessive burn-in requirements
- Persistent initialization bias
- Failure to detect convergence issues

► Theoretical Edge Cases

Improper Priors

Improper priors in Bayesian analysis can lead to improper posterior distributions, violating the algorithm's assumptions.

Non-Ergodic Chains

Non-ergodic chains can occur when:

- The proposal distribution fails to reach certain regions
- The chain is reducible
- The chain is periodic

Infinite Variance Targets

Infinite variance targets violate the conditions for the Central Limit Theorem for ergodic averages, making standard error estimates invalid.

- ▶ **Extremely High Dimensions:** When $d > 100$, the random walk acceptance rate plummets. To maintain a reasonable acceptance rate (e.g., 23.4% for high dimensions), the step size must shrink to near zero. The chain effectively freezes, requiring billions of iterations to move anywhere.
- ▶ **Disconnected Multimodal Distributions:** If the modes are separated by regions where $\pi(x) \approx 0$ (e.g., a Gaussian mixture with far-apart means), the chain cannot cross the gap. The irreducibility assumption holds mathematically (because there is a non-zero path), but it fails practically because the probability of making that path in finite time is effectively zero.

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